

# Microorganisms and Disease — Complete Notes

Grade 11

Life Sciences

CAPS

## What you'll learn

- The four main groups of microorganisms and how to tell them apart
- The basic structure and general characteristics of viruses, bacteria, protists and fungi
- How microorganisms cause disease, and how diseases are transmitted and prevented
- How the human body defends itself, and the four types of immunity
- How vaccines work, step by step
- The role of microorganisms in traditional and modern biotechnology
- How to write full, reasoned answers and do simple calculations on disease data

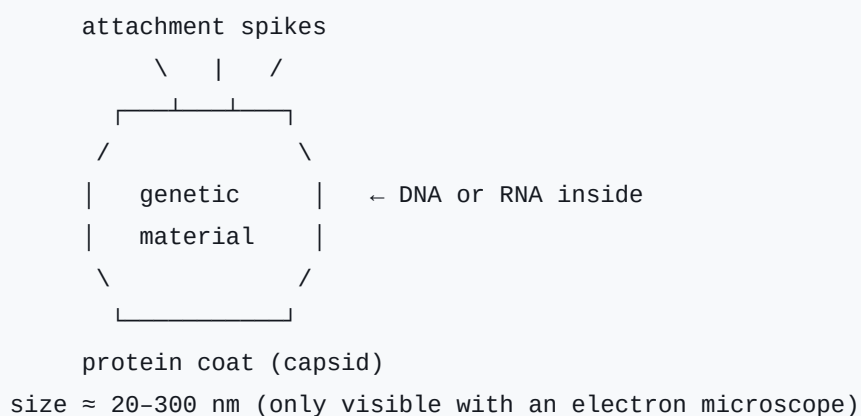
## What is a microorganism?

A **microorganism** (also spelled micro-organism, or simply a **microbe**) is a living organism that is too small to be seen with the naked eye — you need a microscope. Microorganisms live almost everywhere: in soil, water and air, on your skin, and inside your body. Most are harmless or even useful, but a small number cause disease. A microorganism that causes disease is called a **pathogen**.

CAPS groups microorganisms into four main types: **viruses, bacteria, protists (protozoa) and fungi**.

## Group 1: Viruses

Viruses are the odd group out. They are **not** made of cells, so they sit on the borderline between living and non-living.



### General characteristics of viruses

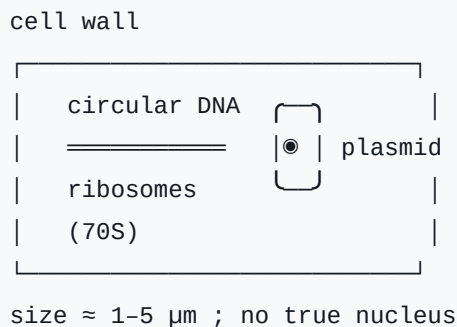
- They have **no cell structure** — no cytoplasm, no nucleus, no cell membrane and no ribosomes.
- They consist of a **protein coat (capsid)** surrounding genetic material, which is either **DNA or RNA** (never both).
- They can only **reproduce inside a living host cell** (they are obligate intracellular parasites). Outside a host they are inactive, like crystals.
- They have **no metabolism** of their own — they use the host cell's enzymes and energy.

- They are **much smaller** than bacteria (nanometres, not micrometres).
- They **mutate quickly**, which is why new strains keep appearing (e.g. new flu strains).
- Examples: HIV (AIDS), influenza, measles, rabies and the common cold viruses.

Because viruses lack the characteristics of living cells, antibiotics do **not** work against them — antibiotics only attack bacteria.

## Group 2: Bacteria

Bacteria are **prokaryotes**: single-celled organisms whose genetic material is **not** enclosed in a nuclear membrane.



### General characteristics of bacteria

- **Unicellular** and **prokaryotic** — there is no nuclear membrane, so the DNA lies free in the cytoplasm.
- Have a **cell wall** (mostly of a substance called peptidoglycan), a cell membrane, cytoplasm and **ribosomes**.
- The genetic material is a single **circular DNA** molecule; many bacteria also carry small rings of DNA called **plasmids**.
- Reproduce **asexually by binary fission** (one cell divides into two) — very fast, e.g. every 20 minutes in good conditions.
- Some form resistant **endospores** that survive harsh conditions.
- Nutrition: most are **heterotrophic**, but some are **autotrophic**.
- Movement: many have tail-like **flagella**.
- Many bacteria are useful (decomposers, nitrogen-fixing bacteria, those used in yoghurt and cheese), but some are pathogens: *Mycobacterium tuberculosis* (TB) and *Vibrio cholerae* (cholera).

## Group 3: Protists (protozoa)

Protists are **eukaryotic** microorganisms — their DNA is enclosed in a true nucleus.

- Mostly **unicellular**, but with a true **nucleus** and membrane-bound organelles.
- **No cell wall** (in protozoa), so many can change shape.
- Move using **pseudopods, cilia or flagella**.
- Most are **heterotrophic**; some are parasites.
- Reproduce mainly by **cell division**.
- Disease-causing examples: *Plasmodium* (causes **malaria**), transmitted by the bite of an infected **female Anopheles mosquito**; also *Trypanosoma* (sleeping sickness, spread by the tsetse fly) and *Entamoeba*

(amoebic dysentery).

## Group 4: Fungi

Fungi are also **eukaryotic** and can be unicellular or multicellular.

- Unicellular (e.g. **yeast**) or multicellular (e.g. **moulds**).
- Cell walls made of **chitin** (not cellulose).
- **Heterotrophic** — they digest food outside the body and absorb it.
- Reproduce **sexually or asexually by spores** — huge numbers of tiny spores are carried by air.
- They have **no chlorophyll**, so they cannot photosynthesise.
- Useful fungi: yeast in brewing and baking, *Penicillium* for antibiotics, and decomposers. Disease-causing: *Candida albicans* (thrush) and dermatophytes (ringworm and athlete's foot).

## How microorganisms cause disease

### Key terms

- **Pathogen** — a microorganism that causes disease.
- **Infection** — when a pathogen enters the body and multiplies.
- **Disease** — the symptoms caused by the infection.
- **Vector** — an organism that carries and transmits a pathogen from one host to another (e.g. a mosquito).
- **Carrier** — a person who carries a pathogen without showing symptoms but can still spread it.

### How pathogens harm the body

- They **use the host's resources** (food, energy), so the body weakens.
- They **damage cells and tissues** (e.g. destroy lung tissue in TB).
- Many produce **toxins** (poisons) that cause fever and other symptoms.
- The body's own **immune response** also causes symptoms such as fever, swelling and pain.

### Transmission and prevention

| Mode of transmission                   | Example disease            | Prevention                                  |
|----------------------------------------|----------------------------|---------------------------------------------|
| Airborne droplets (sneezing, coughing) | Tuberculosis, influenza    | Cover mouth, ventilate rooms, vaccinate     |
| Contaminated water or food             | Cholera, typhoid           | Clean water, sanitation, wash hands         |
| Vector-borne (mosquito, tsetse fly)    | Malaria, sleeping sickness | Nets, repellents, destroy breeding sites    |
| Body fluids (blood, sex)               | HIV/AIDS                   | Safe sex, screened blood, no shared needles |
| Direct contact                         | Ringworm, athlete's foot   | Clean skin, do not share towels             |

### Named diseases to know

| Disease           | Pathogen                          | Type of microorganism | Vector or route                  |
|-------------------|-----------------------------------|-----------------------|----------------------------------|
| Tuberculosis (TB) | <i>Mycobacterium tuberculosis</i> | Bacterium             | Airborne droplets                |
| Malaria           | <i>Plasmodium</i>                 | Protozoan (protist)   | Female <i>Anopheles</i> mosquito |
| HIV/AIDS          | Human immunodeficiency virus      | Virus                 | Body fluids                      |
| Cholera           | <i>Vibrio cholerae</i>            | Bacterium             | Contaminated water               |
| Thrush            | <i>Candida albicans</i>           | Fungus                | Overgrowth on mucous membranes   |

## The body's defences and immunity

The body defends itself in three levels.

- **First line of defence (non-specific, physical and chemical):** intact **skin**, **mucous membranes**, hairs and cilia in the nose, tears and saliva (which contain the enzyme lysozyme), and acid in the stomach.
- **Second line of defence (non-specific, internal):** **inflammation** (redness, heat, swelling), **phagocytes** (macrophages and neutrophils) that engulf pathogens, **fever**, and natural killer cells.
- **Third line of defence (specific):** **lymphocytes**.
- **B-lymphocytes** produce **antibodies** that bind to specific **antigens** (humoral immunity).
- **T-lymphocytes** help B-cells and destroy infected cells (cell-mediated immunity).
- Some lymphocytes become **memory cells**, so a second infection by the same pathogen is fought faster and harder.

## The four types of immunity

| Type               | How it is acquired                                 | Example                             |
|--------------------|----------------------------------------------------|-------------------------------------|
| Natural active     | Own body makes antibodies after an infection       | Recovering from measles             |
| Natural passive    | Antibodies pass from mother to baby                | Across the placenta; in breast milk |
| Artificial active  | Own body makes antibodies after a <b>vaccine</b>   | Measles or TB vaccine               |
| Artificial passive | Antibodies are <b>injected</b> from another source | Anti-venom, tetanus immunoglobulin  |

**Vaccination** works by injecting a weakened, killed or harmless part of a pathogen (or its toxin). The immune system responds by making antibodies and **memory cells** without the person becoming seriously ill. If the real pathogen attacks later, the memory cells trigger a fast secondary response, so the person does not become ill.

## Biotechnology

**Biotechnology** is the use of living organisms (or their parts) to make useful products or provide services.

- **Traditional biotechnology** — used for centuries: yeast in **bread** (CO<sub>2</sub> makes dough rise), **brewing** beer and wine, **Lactobacillus** bacteria in **yoghurt**, **amasi** (sour milk) and cheese, and vinegar production.

- **Modern biotechnology** — uses gene technology (genetic engineering and recombinant DNA):
- Bacteria engineered to produce human **insulin** for diabetics.
- Producing **vaccines** and **antibiotics** (penicillin from *Penicillium*).
- **Gene therapy** — inserting a normal gene into cells with a faulty one.
- Genetically modified (GM) crops with resistance to pests or drought.

## Worked examples

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### Worked example 1 — Identify and justify

A blood sample shows a single-celled pathogen that has a true nucleus, has no cell wall, and is found inside red blood cells. The patient has recurring fevers. Name the group and suggest how it was transmitted. Justify each point.

**Step 1 — Identify the group.** It is a **protist (protozoan)**. The clue "single-celled organism with a true nucleus" means it is **eukaryotic**, but "no cell wall" rules out fungi.

**Step 2 — Identify the likely pathogen.** Living inside red blood cells and causing recurring fever points to ***Plasmodium***, the cause of **malaria**.

**Step 3 — Transmission.** *Plasmodium* is transmitted by the bite of an **\*\*infected female *Anopheles* mosquito, which acts as a vector\*\***.

**Answer:** Protozoan (protist); *Plasmodium*; transmitted by the *Anopheles* mosquito vector.

### Worked example 2 — Explain a process (vaccination)

Explain, step by step, how a vaccine protects a person against a future infection.

**Step 1** — A vaccine containing weakened, dead or harmless parts of the pathogen (**antigens**) is injected.

**Step 2** — The antigen is recognised as foreign by the immune system.

**Step 3** — **B-lymphocytes** produce specific **antibodies** against that antigen, and **T-lymphocytes** help.

**Step 4** — Some lymphocytes become **memory cells** that remain in the body.

**Step 5** — If the real, active pathogen later enters the body, the memory cells recognise it immediately and trigger a **rapid secondary immune response** — large amounts of antibody are made before symptoms develop.

**Conclusion** — The person is now **immune**; this is **artificial active immunity**.

### Worked example 3 — Working with data

At a clinic, 1 200 people were tested for a disease. Of these, 240 tested positive. **(a)** Calculate the percentage that tested positive. **(b)** If a further 360 people were tested and **15%** were positive, how many total positive cases were there?

**Step 1 (a)** — Percentage positive =  $(240 \div 1\,200) \times 100 = \mathbf{20\%}$ .

**Step 2 (b)** — 15% of 360 =  $(15 \div 100) \times 360 = \mathbf{54}$  additional positive cases.

**Step 3** — Total positives =  $240 + 54 = \mathbf{294}$ .

**Answer:** (a) 20% (b) 294 positive cases.

## Common mistakes

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- Calling viruses "bacteria". Viruses are **not cells** and have **no metabolism**; bacteria are prokaryotic cells.
- Saying antibiotics cure viral diseases. Antibiotics work **only on bacteria**.
- Confusing **active** and **passive** immunity. Active means the body makes its own antibodies (lasts long); passive means ready-made antibodies are given (short-lived).
- Forgetting that a **vector** is not the pathogen — the mosquito *carries Plasmodium* but is not the parasite itself.
- Confusing a **carrier** (no symptoms but still spreads the pathogen) with a **vector** (usually another organism).
- Saying protozoa are fungi — protists are **eukaryotic**, have **no cell wall**, and belong to a separate group.
- Forgetting to justify each statement with a reason in "explain" questions.

## Quick summary / cheat sheet

| Group    | Cells?          | Nucleus?        | Cell wall        | Reproduces        | Disease example |
|----------|-----------------|-----------------|------------------|-------------------|-----------------|
| Viruses  | No              | No              | No (capsid only) | Only in host cell | HIV, flu        |
| Bacteria | Yes             | No (prokaryote) | Yes              | Binary fission    | TB, cholera     |
| Protists | Yes             | Yes (eukaryote) | No               | Cell division     | Malaria         |
| Fungi    | Yes/unicellular | Yes (eukaryote) | Yes (chitin)     | Spores            | Thrush          |

- **Immunity:** active = own antibodies (long) · passive = given antibodies (short).
- **Vaccine** → **antigens** → **antibodies + memory cells** → **immunity**.
- **Biotechnology:** traditional (bread, beer, yoghurt) vs modern (insulin, vaccines, gene therapy).
- **Prevention basics:** clean water, sanitation, vaccination, vector control, hygiene.

## Practice questions

1. Give **two** reasons why viruses are said to be on the borderline between living and non-living.
2. Name the microorganism group whose cell wall is made of **chitin**, and give one disease it causes.
3. Explain the difference between **natural** and **artificial** passive immunity, with one example of each.
4. A learner says, "I have flu, so I will take antibiotics." Is this correct? Explain your answer.
5. List **three** first-line defences of the human body against pathogens.
6. In a school of 800 learners, 160 have a cold. Calculate the percentage affected.
7. Explain how **binary fission** allows a bacterial infection to spread so quickly through the body.

## Answers

1. They have no cell structure (no cytoplasm, nucleus or organelles); they have no metabolism of their own and can only reproduce inside a living host cell.
2. **Fungi**; for example thrush (*Candida albicans*), ringworm or athlete's foot.
3. **Natural passive** — antibodies pass from mother to baby naturally (across the placenta or in breast milk). **Artificial passive** — ready-made antibodies are injected (e.g. anti-venom or tetanus immunoglobulin).
4. Incorrect. Flu is caused by a **virus**, not a bacterium. Antibiotics only act on bacteria, so they will not kill the flu virus.

5. Any three of: intact skin; mucous membranes; cilia and hairs in the nose; tears and saliva (lysozyme); stomach acid.
6.  $(160 \div 800) \times 100 = \mathbf{20\%}$ .
7. In binary fission one bacterium divides into two identical cells, and each of those divides again, so the number can double roughly every 20 minutes. This exponential growth quickly produces a huge number of pathogens in the body.

Official EduRank Team notes. Always cross-check with your textbook and teacher.